The Toy Workshop Theft

Whodunnit? · Similarity · Teacher Guide and Answer Key

Unit	Time	Supplies	Group size
Similarity	20–40 minutes	Pencil, scrap paper, scissors	Teams of 2–3 (or solo)

Skills practiced: Similar figures: scale factors, proportions, perimeter and area ratios, indirect measurement.

General procedures

1. Give each student the **student sheet** (story and case board) and the **Show Your Work** page.
2. Print the **Clue Cards** (pages 3–7 of the student packet) and cut them apart on the dashed lines. Laminating them lets you reuse the set. Post them around the room.
3. Students visit the cards in any order. At each card they solve the problem on their work page, find their answer among the four choices, read the clue beside it, and cross things off the case board. **Every wrong choice leads to a false clue** that clears the real culprit, so a mistake shows up when the board does not narrow down to one suspect, one place and one method.
4. When a team thinks it knows, they bring you the three names. If they are wrong, let them hunt for the card where something went wrong.

Solution: Ben Tran took it, in the Testing Lab, using this method: Hidden in a teddy bear.

The reveal (read aloud): During the unveiling, **Ben Tran** slipped into the **Testing Lab**, stuffed Tiny Titan inside a **teddy bear**, and walked out hugging it like a toy. Every alibi and method you cleared led straight to him. A perfect fit!

Clue map: what each correct clue clears

Card	Clears
1	Suspect: Mr. Gearhart · Place: Design Studio
2	Method: Packed in a toy box
3	Suspect: Ms. Pom-Pom · Place: Paint Room
4	Method: Rolled out on a skateboard
5	Suspect: Dr. Doodle · Place: Assembly Line
6	Method: Dressed as a doll
7	Suspect: Mrs. Wick · Place: Packing Dock
8	Method: Shipped in a gift box
9	Suspect: Mr. Spoole · Place: Teddy Corner
10	Method: Stashed in a shopping bag

Answer Key

The Toy Workshop Theft · Correct choice and worked solution for each card

#	Problem	Correct choice	Worked solution	Clue it leads to
1	△ABC ~ △DEF. AB = 4, BC = 6 and DE = 6. Find EF.	**A.** 9	$\frac{4}{6} = \frac{6}{EF}$ EF = 9	Mr. Gearhart was sketching designs in the Design Studio. The late-night guard saw him.
2	Solve the proportion: $\frac{x}{15} = \frac{4}{10}$	**D.** 6	10x = 60 x = 6	Every toy box was opened and emptied. No Tiny Titan.
3	A triangle with a side of 6 is enlarged to a similar triangle with the matching side 9. Find the scale factor.	**C.** $\frac{3}{2}$	scale factor = $\frac{new}{old} = \frac{9}{6}$ $\frac{3}{2}$	Ms. Pom-Pom was painting a rocking horse in the Paint Room. Wet paint all over her sleeves proves it.
4	Two similar polygons have a scale factor of 2 : 5. What is the ratio of their perimeters?	**A.** $\frac{2}{5}$	perimeters have the same ratio as the sides $\frac{2}{5}$	The skateboard in the hall has a flat wheel. It has not rolled in days.
5	Two similar polygons have a scale factor of 2 : 5. What is the ratio of their areas?	**A.** $\frac{4}{25}$	areas use the square of the scale factor: $\frac{2^2}{5^2}$ $\frac{4}{25}$	Dr. Doodle was testing wind-up cars on the Assembly Line with three other testers.
6	A 6-ft person casts a 4-ft shadow. A tree casts a 14-ft shadow at the same time. How tall is the tree?	**B.** 21 ft	$\frac{6}{4} = \frac{h}{14}$ 4h = 84 21 ft	The dolls on the shelf are standing in neat rows, each one a normal-sized doll.
7	Rectangle A is 3 by 5. Rectangle B is similar with a short side of 9. Find the long side of Rectangle B.	**D.** 15	scale factor = $\frac{9}{3} =$ 3 5 · 3 15	Mrs. Wick was taping boxes at the Packing Dock, timed by the dock camera.
8	On a map, 1 inch represents 25 miles. Two cities are 3.5 inches apart on the map. How many miles apart are they?	**B.** 87.5 mi	3.5 × 25 87.5 mi	The gift boxes were opened and counted. All held the gifts they should.
9	In △ABC, DE ∥ BC with D on AB and E on AC. AD = 4, DB = 6 and AE = 5. Find EC.	**C.** 7.5	$\frac{AD}{DB} = \frac{AE}{EC}$ $\frac{4}{6} = \frac{5}{EC}$ 4 · EC = 30 7.5	Mr. Spoole was reading stories to kids in Teddy Corner for the whole hour.
10	Two similar figures have areas 18 and 50. Find the scale factor of the smaller to the larger.	**B.** $\frac{3}{5}$	ratio of areas = $\frac{18}{50} = \frac{9}{25}$ scale factor = √$\frac{9}{25}$ $\frac{3}{5}$	The shopping bags were checked at the door. Only lunches and umbrellas.

Wrong-Answer Map

The Toy Workshop Theft · What a wrong choice looks like, and the false clue it leads to

Card	Wrong choice	Likely error	False clue
1	B. 10	Added 4 to 6	Every teddy bear on the shelf was squeezed and shaken. Nothing hard inside any of them.
1	C. 4	Inverted the scale factor	Ben Tran was sharpening pencils in the Design Studio. Mr. Gearhart said he stayed all afternoon.
1	D. 8	Added the difference instead of multiplying	Mr. Gearhart was checking the gears of a toy dog in the Testing Lab. The tester signed his visit.
2	A. 60	Multiplied 15 by 4 without dividing	The teddy bears were all sewn shut with tight factory stitches. None had been torn open.
2	B. 9	Subtracted instead of using proportion	Ms. Pom-Pom was testing a rainbow robot in the Testing Lab. The lab log lists her name.
2	C. 37.5	Inverted the proportion	Ben Tran was mixing paint in the Paint Room. Blue splotches cover his apron.
3	A. 3	Subtracted $9 - 6$	Mrs. Wick was wrapping a prize in the Testing Lab. The testers gave her a round of applause.
3	B. $\frac{6}{15}$	Divided 6 by 6 + 9	The teddy bears are too small and floppy to hold a robot. Nobody could hide Tiny Titan in one.
3	D. $\frac{2}{3}$	Inverted the scale factor	Ben Tran was oiling gears on the Assembly Line. The foreman watched him the whole shift.
4	B. $\frac{8}{125}$	Cubed the ratio (volume ratio)	Mr. Spoole was dropping off a package at the Testing Lab. The sign-in sheet has his name.
4	C. $\frac{5}{2}$	Inverted the ratio	Teddy Corner’s bears were counted at closing. Every bear was accounted for and untouched.
4	D. $\frac{4}{25}$	Squared the ratio (area ratio)	Ben Tran was loading pallets at the Packing Dock. The dock camera caught him every minute.
5	B. $\frac{25}{4}$	Inverted the area ratio	The teddy bear with the torn ear was checked. Only stuffing inside.
5	C. $\frac{8}{125}$	Cubed the ratio	Dr. Doodle was timing a toy race in the Testing Lab. The stopwatch still shows the time.
5	D. $\frac{2}{5}$	Used the unsquared ratio	Ben Tran was fluffing pillows in Teddy Corner. The kids giggled as he worked.
6	A. $\frac{28}{3}$ ft	Inverted the proportion	Mr. Gearhart was tightening bolts on a toy crane in the Testing Lab. His wrench is still there.
6	C. 18 ft	Added 4 to 14	The bears were scanned at the gate. The pictures show only fluff.
6	D. 10 ft	Subtracted 4 from 14	Ben Tran was drawing blueprints in the Design Studio. The ink was still wet.
7	A. 5.4	Inverted the scale factor	Ben Tran was washing brushes in the Paint Room. The sink is still full of colorful water.
7	B. 11	Added 2 to 9	Ms. Pom-Pom was handing out stickers in the Testing Lab. The kids all remember her.
7	C. 27	Multiplied 9 by 3 twice	The teddy bear on the night shelf has a tag dated yesterday. It was never opened.
8	A. 875 mi	Misplaced the decimal	Every bear’s tummy zipper was found zipped and locked. No one had opened a single one.

8	C. 28.5 mi	Added 25 and 3.5	Mrs. Wick was delivering boxes to the Testing Lab. The delivery slip is stamped.
8	D. 0.14 mi	Divided 3.5 by 25	Ben Tran was fixing a conveyor belt on the Assembly Line with two other workers.
9	A. 9	Added 4 to 5	The teddy bears were lined up and counted at the door. Not one was missing or lumpy.
9	B. $\frac{10}{3}$	Inverted the proportion	Mr. Spoole was waiting outside the Testing Lab for a signature. The clerk kept him company.
9	D. 12.5	Found AC instead of EC	Ben Tran was sweeping the Packing Dock. The pile of packing peanuts is proof.
10	A. $\frac{9}{25}$	Used the area ratio	Ben Tran was reading a picture book in Teddy Corner. The kids sat all around him.
10	C. $\frac{5}{3}$	Inverted the scale factor	Dr. Doodle was drop-testing a toy truck in the Testing Lab. The dents prove it.
10	D. $\frac{25}{9}$	Inverted the area ratio	The big teddy bear by the door was checked. Its stuffing was fluffy and fresh.